HW #5

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1.

(a) The "primitive soup" hypothesis (Oparin-Haldane) says life began in a the ocean where UV light and lightning produced organic compounds like amino acids and nucleotides. The "soup" has no way to ensure that polymers form with the correct chemical bonds. Proteins require peptide bonds and RNA requires specific nucleotide linkages, but Miller-type experiments produce a mix of bonds. Wrong linkages block replication entirely.

The "primitive pizza" hypothesis (Wachtershauser) says life began on positively charged iron pyrite surfaces. Organic ions bond to the surface and are held in a fixed orientation, able to move only in two dimensions instead of freely in solution. This increases both reaction speed and specificity, and concentrates molecules so they can react. This directly solves the soup's problem because the surface enforces correct orientations for polymerization.

(b) Without replicase enzymes the error rate is very high, too high to copy any meaningful genetic sequence reliably. With replicase but without proofreading, the error rate drops to 1/1000 to 1/10,000 per nucleotide. RNA strands separate immediately after copying so there is no way to check for mistakes, unlike DNA which uses the second strand to catch errors. Replicase helps but proofreading is what really makes large genomes viable.

(c) RNA is single-stranded so errors cannot be proofread. With a human genome of 3 billion bases and an error rate of 1/1000 to 1/10,000, every replication cycle would introduce thousands to millions of mutations, far past the point where useful genetic information survives. RNA also degrades faster chemically than DNA, making long-term storage of a complex genome impossible.

2.

(a)

$$m_{\text{proton}} = m_{\text{neutron}} = 1.67 \times 10^{-27} kg$$
$$m_{^{12}\text{C}} = 12 \times 1.67 \times 10^{-27} \text{ kg} = 2.004 \times 10^{-26} \text{ kg}$$
$$m_{^{13}\text{C}} = 13 \times 1.67 \times 10^{-27} \text{ kg} = 2.171 \times 10^{-26} \text{ kg}$$

(b) Using $E_1 = \frac{h^2}{8mL^2}$ with $L = 2 \times 10^{-10}$ m and $h = 6.626 \times 10^{-34}$ J s:

$$E_1(^{12}\text{C}) = \frac{(6.626 \times 10^{-34})^2}{8 \times 2.004 \times 10^{-26} \times (2 \times 10^{-10})^2} = 6.85 \times 10^{-23} \text{ J} = \frac{6.85 \times 10^{-23}}{1.602 \times 10^{-19}} \times 10^3 \text{ meV} \approx \boxed{0.427 \text{ meV}}$$

$$E_1(^{13}\text{C}) = \frac{(6.626 \times 10^{-34})^2}{8 \times 2.171 \times 10^{-26} \times (2 \times 10^{-10})^2} = 6.32 \times 10^{-23} \text{ J} = \frac{6.32 \times 10^{-23}}{1.602 \times 10^{-19}} \times 10^3 \text{ meV} \approx \boxed{0.394 \text{ meV}}$$

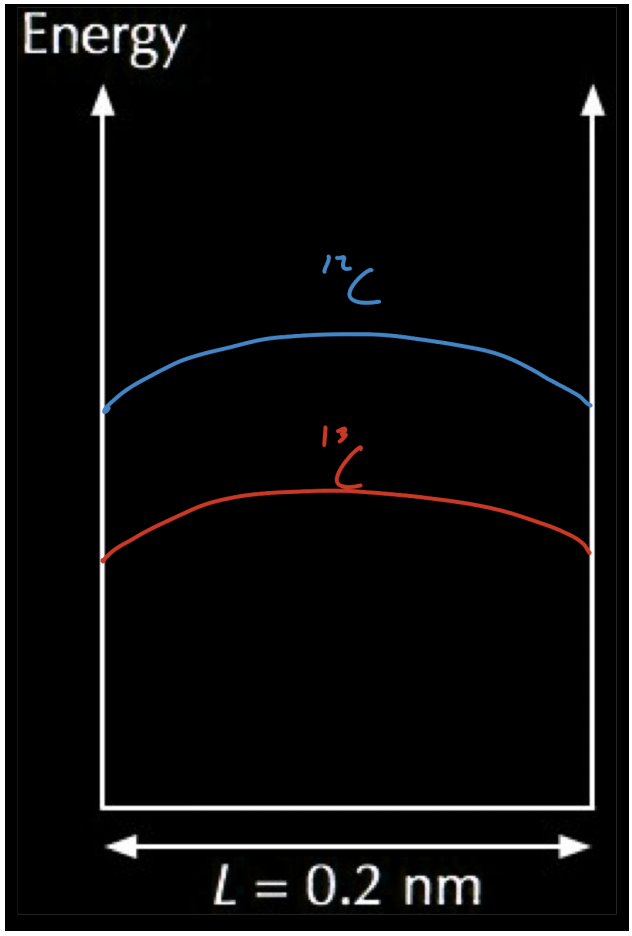

Figure 1: Ground-state ($n = 1$) wavefunctions for ^{12}C (blue) and ^{13}C (red). Because ^{12}C is lighter, it has a higher energy.

(c) Both isotopes have the same transition-state energy E_{ts}, so their activation energies are $E_a = E_{\text{ts}} - E_1$. Since ^{12}C has a higher E_1, it has a lower activation energy and reacts faster. The E_{ts} terms cancel in the ratio:

$$\frac{\text{rate}(^{12}\text{C})}{\text{rate}(^{13}\text{C})} = e^{(E_1^{12} - E_1^{13})/k_B T}$$

$$\Delta E = 6.85 \times 10^{-23} - 6.32 \times 10^{-23} = 5.3 \times 10^{-24}\ \text{J}, \qquad k_B T = (1.381 \times 10^{-23})(300) = 4.14 \times 10^{-21}\ \text{J}$$

$$\frac{\text{rate}(^{12}\text{C})}{\text{rate}(^{13}\text{C})} = e^{5.3\times10^{-24}/4.14\times10^{-21}} = e^{1.28\times10^{-3}} \approx \boxed{1.00128}$$

^{12}C reacts about 0.13% faster, consistent with observed isotope fractionation in photosynthesis.

(d) At higher temperature, thermal energy $k_B T$ grows, so the energy difference ΔE becomes relatively smaller. The rate ratio approaches 1, so both isotopes react at nearly the same rate.

(e) For ^{13}C to react faster, it would need a lower activation energy than ^{12}C. Since ^{12}C already has higher ground-state energy, this would require the transition state to also have zero-point energy, and for the ^{12}C transition-state energy to be even higher relative to the difference in their ground states. This would happen if the carbon bond is stiffer at the transition state than at the reactant — giving ^{12}C a larger energy cost there.

3.

(a) Primitive fatty acid membranes are permeable to nucleotides, so they can cross in and out freely. Inside the vesicle, RNA replication consumes free nucleotides by incorporating them into long polymers that cannot leave. This lowers the internal nucleotide concentration below the outside, creating a gradient that drives more nucleotides in than out.

(b) RNA polymers trapped inside raise the total solute concentration inside the vesicle. Water then flows in by osmosis, trying to equalize the concentration across the membrane, causing the vesicle to swell.

(c) Swollen vesicles have membrane tension, which is lowered when new fatty acids are added to the membrane. This means swollen, RNA-rich vesicles preferentially absorb fatty acids from the surrounding solution and from relaxed, RNA-poor neighbors, which then shrink. Better replicators swell more, steal more membrane, and grow at the expense of poorer replicators – natural selection at the protocell level.

(d) Szostak proposes temperature cycling driven by volcanic environments. Protocells in a pool near hot rocks would occasionally be carried by convection currents past the rocks, receiving a brief burst of heat that separates the two RNA strands. They then cool rapidly in the bulk cold water, where each single strand acts as a template to form a new complementary strand. This is essentially a natural geothermal version of PCR, requiring no molecular machinery.

4.

(a) The L-form is favored by $\Delta E = 10^{-17}$ eV, so the probability ratio is:

$$\frac{P(L)}{P(D)} = e^{\Delta E/k_B T}, \qquad k_B T = (8.617 \times 10^{-5})(295) = 0.02542\ \text{eV}$$

$$\frac{P(L)}{P(D)} = e^{10^{-17}/0.02542} = e^{3.93\times10^{-16}} \approx \boxed{1 + 3.93 \times 10^{-16}}$$

The two forms are essentially equally probable at room temperature. The L-form advantage is only 1 part in 10^{16}.

(b) For $P(D)/P(L) = 0.1$:

$$e^{-\Delta E/k_B T} = 0.1 \implies -\frac{\Delta E}{k_B T} = \ln(0.1) = -2.303 \implies k_B T = \frac{\Delta E}{2.303} = \frac{10^{-17}}{2.303} = 4.34 \times 10^{-18} \text{ eV}$$

$$T = \frac{4.34 \times 10^{-18}}{8.617 \times 10^{-5}} \approx \boxed{5.04 \times 10^{-14} \text{ K}}$$

This is far below anything achievable in nature.

(c) No. The effect at room temperature is only on the order 10^{-16}, which is nowhere near enough to explain how there is near 100% L-amino acid homochirality in biology.

(d) The alternative explanation is that homochirality was a historical accident. Abiotic chemistry produces a 50/50 mix of L and D amino acids, but by chance one form ended up slightly more prevalent early in the origin of life. Once one handedness had a small advantage, it became locked in and amplified over time since mixing L and D disrupts protein function.